

```

clc; clear

Fs=10000; Fp=3000; Fst=2000;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn,'high');

% Magnitude & Phase
freqz(b,a)

% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')

% Step Response
figure
stepz(b,a)
title('Step Response')

% Group Delay
figure
grpdelay(b,a)
title('Group Delay')

% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);

```

